

SUPERSTORE SALES BUSINESS ANALYSIS

SQL Query Report

Portfolio Project

Analyzed 9,800 Superstore records from 2015–2018 to evaluate regional, category, customer, monthly sales and shipping performance.

Original SQL code & result screenshots preserved

Superstore Sales sql queries



Objectives

The goal of this analysis was to identify top-performing and underperforming areas across product categories, regions, states, and customer segments using the Superstore sales dataset(2015–2018) of 9,800 rows. The objective was to uncover actionable business insights that can improve profitability, optimize product strategy, and support data-driven decision making.

QUERIES FOR KPI'S AND CHECKING THE DATA:

(" is for ")

Query :

```
-- total number of coustomer we have ?  
SELECT COUNT(DISTINCT "customer id") AS total_customers  
FROM superstore_db;
```

- By this query we are able to find total number of customers in 9,800 rows of data.

Total_customers
793

Query :

```
-- type of sub_category we are selling?  
select "Sub-Category" from superstore_db group by "Sub-Category";
```

- By this we will able to find how many sub-category exists in our FURNITURE , OFFICE SUPPLIES , TECHNOLOGies products in our store.

	Total_sub_Category
1	Accessories
2	Appliances
3	Art
4	Binders
5	Bookcases
6	Chairs
7	Copiers
8	Envelopes
9	Fasteners
10	Furnishings
11	Labels
12	Machines
13	Paper
14	Phones
15	Storage
16	Supplies
17	Tables

Query :

```
-- checking if we have any null values rows
SELECT *
FROM superstore_db
WHERE "Order ID" IS NULL
    OR "Order Date" IS NULL
    OR "Ship Date" IS NULL
    OR "Ship Mode" IS NULL
    OR "Customer ID" IS NULL
    OR "Customer Name" IS NULL
    OR "Segment" IS NULL
    OR "Country" IS NULL
    OR "City" IS NULL
    OR "State" IS NULL
    OR "Postal Code" IS NULL
    OR "Region" IS NULL
    OR "Product ID" IS NULL
    OR "Category" IS NULL
    OR "Sub-Category" IS NULL
    OR "Product Name" IS NULL
    OR Sales IS NULL;
```

•By this query we are able to find if there is any missing data and in our data there is POSTAL CODES for Burlington in east region are missing.

	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Na
1	QJ-19255	Quincy Jones	Corporate	United States	Burlington	Vermont	NULL	East	TEC-AC-10001013	Technology	Accessories	Logitech ClearChat Comfort/USB H
2	SV-20785	Stewart Visinsky	Consumer	United States	Burlington	Vermont	NULL	East	FUR-CH-10000595	Furniture	Chairs	Safco Contoured Stacking Chairs
3	VM-21685	Valerie Mitchum	Home Office	United States	Burlington	Vermont	NULL	East	TEC-PH-10002555	Technology	Phones	Nortel Meridian M5316 Digital phon
4	CB-12535	Claudia Bergmann	Corporate	United States	Burlington	Vermont	NULL	East	TEC-AC-10002926	Technology	Accessories	Logitech Wireless Marathon Mouse
5	CB-12535	Claudia Bergmann	Corporate	United States	Burlington	Vermont	NULL	East	OFF-AR-10003477	Office Supplies	Art	4009 Highlighters
6	CB-12535	Claudia Bergmann	Corporate	United States	Burlington	Vermont	NULL	East	OFF-ST-10001526	Office Supplies	Storage	Iceberg Mobile Mega Data/Printer C
7	RM-19375	Raymond Messe	Consumer	United States	Burlington	Vermont	NULL	East	OFF-PA-10000157	Office Supplies	Paper	Xerox 191
8	RM-19375	Raymond Messe	Consumer	United States	Burlington	Vermont	NULL	East	OFF-PA-10001970	Office Supplies	Paper	Xerox 1881
9	RM-19375	Raymond Messe	Consumer	United States	Burlington	Vermont	NULL	East	OFF-AP-10000828	Office Supplies	Appliances	Avanti 4.4 Cu. Ft. Refrigerator
10	RM-19375	Raymond Messe	Consumer	United States	Burlington	Vermont	NULL	East	OFF-EN-10001509	Office Supplies	Envelopes	Poly String Tie Envelopes
11	QJ-19255	Quincy Jones	Corporate	United States	Burlington	Vermont	NULL	East	FUR-BO-10004834	Furniture	Bookcases	Riverside Palais Royal Lawyers Boc

Queries to get business insight of data:

Query 1:

```
--1 what are the total sales and profit by region?  
  
select region, sum(sales) as total_sales_byregion , sum(sales)* 25/100 as profit  
from superstore_db GROUP by region order by profit desc;  
--assuming we always get 25% of profit from total Sales
```

- The query compares total sales across regions. The original calculation assumes a 25% margin, so the result should be treated as estimated profit, not actual profit.

Region	Total_sales_byregion	Profit
West	710219.6845	177554.921125
East	669518.726	167379.6815
Central	492646.9132	123161.7283
South	389151.459	97287.86475

- West generates the highest sales, while South generates the lowest. This can guide deeper analysis of regional demand, category mix and market performance.

Query 2:

```
--2. What is the total REVENUE(In OUR CASE its SALES) of each Category?  
SELECT Category, SUM(Sales) AS TOTAL_REVENUE FROM superstore_db  
GROUP BY Category ORDER BY TOTAL_REVENUE DESC;
```

- The current query compares every category within each region, it does not select only the top category.
- We can Use this comparison to identify leading and lagging categories , can be margins before expanding or reducing a category.

	Category	TOTAL_REVENUE
1	Technology	827455.873
2	Furniture	728658.5757
3	Office Supplies	705422.334

Query 3:

```
--3. WHICH Category SELLING MOST in which region?  
SELECT Region, Category, SUM(Sales) AS Total_Sales  
FROM superstore_db  
GROUP BY Region, Category  
ORDER BY Region, Total_Sales DESC;
```

- Technology generates the highest total sales among the categories.
- Use this as a starting point for product-level, regional and customer-segment analysis before changing product.

	Region	Category	Total_Sales
1	Central	Technology	168739.208
2	Central	Office Supplies	163590.243
3	Central	Furniture	160317.4622
4	East	Technology	263116.527
5	East	Furniture	206461.388
6	East	Office Supplies	199940.811
7	South	Technology	148195.208
8	South	Office Supplies	124424.771
9	South	Furniture	116531.48
10	West	Technology	247404.93
11	West	Furniture	245348.2455
12	West	Office Supplies	217466.509

Query 4:

```
--4. WHO ARE THE TOP 10 Customer BY SALES ?  
SELECT "Customer name" , sum(sales) as top_10_sales from superstore_db  
group by "Customer name" order by top_10_sales DESC limit 10 ;
```

- This query identifies the top 10 customers by sales value.

- These high-value customers can be analyzed for purchase frequency, recency and product preferences to support retention and targeted offers.

	Customer Name	top_10_sales
1	Sean Miller	25043.05
2	Tamara Chand	19052.218
3	Raymond Buch	15117.339
4	Tom Ashbrook	14595.62
5	Adrian Barton	14473.571
6	Ken Lonsdale	14175.229
7	Sanjit Chand	14142.334
8	Hunter Lopez	12873.298
9	Sanjit Engle	12209.438
10	Christopher Conant	12129.072

Query 5:

```
SELECT CASE SUBSTR(&quot;ORDER DATE&quot;;4,2)
WHEN '01' THEN 'JAN'
WHEN &quot;02&quot; THEN &quot;FEB&quot;;
WHEN &quot;03&quot; THEN &quot;MARCH&quot;;
WHEN &quot;04&quot; THEN &quot;APR&quot;;
WHEN &quot;05&quot; THEN &quot;MAY&quot;;
WHEN &quot;06&quot; THEN &quot;JUN&quot;;
WHEN &quot;07&quot; THEN &quot;JULY&quot;;
WHEN &quot;08&quot; THEN &quot;AUG&quot;;
WHEN &quot;09&quot; THEN &quot;SEP&quot;;
WHEN &quot;10&quot; THEN &quot;OCT&quot;;
WHEN &quot;11&quot; THEN &quot;NOV&quot;;
WHEN &quot;12&quot; THEN &quot;DEC&quot;;
END AS MONTH, SUM(SALES) AS TOTAL_SALES
FROM superstore_db WHERE &quot;order date&quot;; like '%2017'
GROUP BY SUBSTR(&quot;ORDER DATE&quot;; , 4,2) ORDER BY SUBSTR(&quot;ORDER DATE&quot;; , 4,2);
```

- The query shows monthly sales for 2017 and helps visualize the year's sales pattern.

- The monthly pattern can be used to investigate seasonality and year-over-year changes for future sales planning.

	Customer Name	top_10_sales
1	Sean Miller	25043.05
2	Tamara Chand	19052.218
3	Raymond Buch	15117.339
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Query 6:

```
SELECT CASE SUBSTR(&quot;ORDER DATE&quot;; , 4, 2)
WHEN &quot;01&quot;; THEN &quot;JAN&quot;;
WHEN &quot;02&quot;; THEN &quot;FEB&quot;;
WHEN &quot;03&quot;; THEN &quot;MAR&quot;;
WHEN &quot;04&quot;; THEN &quot;APR&quot;;
WHEN &quot;05&quot;; THEN &quot;MAY&quot;;
WHEN &quot;06&quot;; THEN &quot;JUN&quot;;
WHEN &quot;07&quot;; THEN &quot;JUL&quot;;
WHEN &quot;08&quot;; THEN &quot;AUG&quot;;
WHEN &quot;09&quot;; THEN &quot;SEP&quot;;
WHEN &quot;10&quot;; THEN &quot;OCT&quot;;
WHEN &quot;11&quot;; THEN &quot;NOV&quot;;
WHEN &quot;12&quot;; THEN &quot;DEC&quot;;
END AS MONTHS ,
SUM(CASE WHEN SUBSTR(&quot;ORDER DATE&quot;;, 7,4)= &quot;2015&quot;; THEN SALES END ) AS SALES_2015,
SUM(CASE WHEN SUBSTR(&quot;ORDER DATE&quot;;, 7,4)= &quot;2016&quot;; THEN SALES END) AS SALES_2016,
SUM(CASE WHEN SUBSTR(&quot;ORDER DATE&quot;;, 7,4)= &quot;2017&quot;; THEN SALES END) AS SALES_2017,
SUM(CASE WHEN SUBSTR(&quot;ORDER DATE&quot;;, 7,4)= &quot;2018&quot;; THEN SALES END) AS SALES_2018
from superstore_db
group by substr(&quot;ORDER DATE&quot;; , 4, 2)
order by substr(&quot;ORDER DATE&quot;; , 4, 2) ;
```

- Comparing the same month across years helps identify seasonality, growth patterns and changes in demand.
- These patterns can support inventory, promotional and sales planning during high- and low-performing periods.

	MONTHS	SALES_2015	SALES_2016	SALES_2017	SALES_2018
1	JAN	14206.707	18066.9676	18542.491	43476.474
2	FEB	4619.892	11961.411	22978.815	19920.9974
3	MAR	55205.797	32339.3184	51165.059	58863.4128
4	APR	27906.855	34154.4685	38679.767	35541.9101
5	MAY	23644.303	29959.5305	56656.908	43825.9822
6	JUN	34322.9356	23599.374	39724.486	48190.7277
7	JUL	33781.543	28608.259	38320.783	44825.104
8	AUG	27117.5365	36818.3422	30542.2003	62837.848
9	SEP	81623.5268	63133.606	69193.3909	86152.888
10	OCT	31453.393	31011.7375	59583.033	77448.1312
11	NOV	77907.6607	75249.3995	79066.4958	117938.155
12	DEC	68167.0585	74543.6012	95739.121	83030.3888

Query 7:

```
SELECT YEAR, MONTH, MONTHLY_SALES,  
AVG(MONTHLY_SALES) OVER(PARTITION BY YEAR) AS AVG_YEAR_SALES  
FROM( SELECT SUBSTR(&quot;ORDER DATE&quot;;, 7, 4) AS Year , SUBSTR(&quot;ORDER DATE&quot;;,4,2) AS MONTH,  
SUM(SALES) AS MONTHLY_SALES FROM superstore_db  
GROUP BY SUBSTR(&quot;ORDER DATE&quot;;, 7, 4),SUBSTR(&quot;ORDER DATE&quot;;,4,2));
```

- The query compares each month's sales with the average monthly sales for its own year.

Year	MONTH	MONTHLY_SALES	AVG_YEAR_SALES
2015	01	14206.707	39988.0173416667
2015	02	4519.592	39988.0173416667
2015	03	55206.797	39988.0173416667
2015	04	27906.655	39988.0173416667
2015	05	23644.303	39988.0173416667
2015	06	34322.9356	39988.0173416667
2015	07	33781.543	39988.0173416667
2015	08	27117.5365	39988.0173416667
2015	09	81623.5268	39988.0173416667
2015	10	31453.393	39988.0173416667
2015	11	77907.6607	39988.0173416667
2015	12	68167.0585	39988.0173416667
2016	01	18066.9576	38286.3337833333
2016	02	11951.411	38286.3337833333
2016	03	32339.5184	38286.3337833333
2016	04	34154.4685	38286.3337833333
2016	05	29959.5305	38286.3337833333
2016	06	23599.374	38286.3337833333
2016	07	22606.259	38286.3337833333
2016	08	36818.3422	38286.3337833333
2016	09	63133.606	38286.3337833333
2016	10	31011.7375	38286.3337833333
2016	11	75249.3995	38286.3337833333
2016	12	74543.6012	38286.3337833333
2017	01	18542.491	50016.0458333333
2017	02	22978.815	50016.0458333333
2017	03	81166.069	50016.0458333333
2017	04	35679.767	50016.0458333333
2017	05	56656.908	50016.0458333333
2017	06	39724.485	50016.0458333333
2017	07	38320.783	50016.0458333333
2017	08	30542.2003	50016.0458333333
2017	09	69193.3909	50016.0458333333
2017	10	59583.033	50016.0458333333
2017	11	79066.4958	50016.0458333333
2017	12	95739.121	50016.0458333333
2018	01	43476.474	60171.0016
2018	02	19920.9974	60171.0016

2018	03	58863.4128	60171.0016
2018	04	35541.9101	60171.0016
2018	05	43825.9822	60171.0016
2018	06	48190.7277	60171.0016
2018	07	44825.104	60171.0016
2018	08	62837.848	60171.0016
2018	09	66152.888	60171.0016
2018	10	77448.1312	60171.0016
2018	11	117938.155	60171.0016
2018	12	83030.3888	60171.0016

- This helps identify months that underperform or outperform their yearly baseline and can highlight recurring seasonal patterns.

Query 8:

```
SELECT "Ship Mode",  
       COUNT("Order ID") AS Orders,  
       ROUND(SUM(Sales), 2) AS Total_Sales  
FROM superstore_db  
GROUP BY "Ship Mode";  
ORDER BY Total_Sales DESC;
```

- Standard Class has the highest order volume and sales in the displayed results.

- This can

inform

shipping

capacity and

delivery

planning, but

cost per order, delivery time and

customer satisfaction should be analyzed before

changing the shipping.

	Ship Mode	Orders	Total_Sales
1	Standard Class	5859	1340831.31
2	Second Class	1902	449914.18
3	First Class	1501	345572.26
4	Same Day	538	125219.04

Query 9:

```
SELECT *from(  
SELECT city, &quot;ship mode&quot;, count(*) as orders ,  
row_number() over(PARTITION by city order by count(*) desc) as top_shipmode  
from superstore_db  
where city in ( select city from superstore_db group by city order by sum(sales)  
desc limit 10)  
group by city , &quot;Ship mode&quot;;  
) where top_shipmode = 1;
```

- This analysis identifies the most-used shipping mode among the top 10 cities by sales, helping the business understand

shipping

demand in its

highest-value

markets.

- Standard Class

city	ship mode	orders	top_shipmode
Chicago	Standard Class	188	1
Detroit	Standard Class	69	1
Houston	Standard Class	219	1
Jacksonville	Standard Class	73	1
Los Angeles	Standard Class	438	1
New York City	Standard Class	536	1
Philadelphia	Standard Class	339	1
San Diego	Standard Class	116	1
San Francisco	Standard Class	279	1
Seattle	Standard Class	248	1

is the dominant shipping mode across all 10 cities, suggesting that shipping capacity and operational planning should prioritize this mod